Data Mining Report - I

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1 Introduction

Cancers are among the most significant global health challenges, and effective detection is crucial for successful treatment. Traditional diagnostic methods can be costly, time-consuming, and sometimes yield inconclusive results. As a result, there is growing interest in applying biological data analysis and machine learning techniques to support the diagnosis of cancerous diseases.

This report focuses on data related to prostate cancer. The analysis is based on a DNA microarray dataset containing expression levels of thousands of genes. Such high-dimensional data is potentially valuable for medical diagnostics, as it can contribute to the identification of relevant genetic markers. Moreover, this type of analysis offers insights into functional genomics, helping to uncover the underlying biological mechanisms of the disease.

In this project, data mining techniques were applied to conduct a comprehensive analysis of a selected dataset related to a specific real-world problem. The study includes data visualizations that help illustrate patterns and distributions within the dataset. Furthermore, classification methods were employed, accompanied by a detailed evaluation of model accuracy, providing insights into the effectiveness and reliability of the applied approaches.

2 Dataset introduction

The dataset contains gene expression profiles for 102 prostate samples, including 52 tumor and 50 non-tumor tissues, obtained using Affymetrix microarray technology. The data are normalized, thresholded, and filtered to retain only genes showing substantial variability—specifically, those with at least a fivefold relative change or a minimum of 500 absolute units of difference across samples. This results in a dataset with expression levels for 6,033 genes. The values are presented on a base-10 logarithmic scale and have been standardized per experiment to have zero mean and unit variance across genes.

3 Descriptive analysis

3.1 Overall statistics

Table 1 summarizes key descriptive statistics for the full dataset as well as for the cancerous and non-cancerous groups separately. Due to prior standardization, the overall mean gene expression is centered at zero, with a relatively low standard deviation around 0.24. When examined by class, both groups display comparable central tendencies, though the non-cancerous samples exhibit slightly greater dispersion, reflected in a higher standard deviation and median. This suggests marginally increased heterogeneity in gene expression among non-cancerous tissues. Notably, the Pearson correlation coefficient between the two classes is exceptionally high (0.95), indicating that gene expression profiles are largely consistent across the dataset, with only a limited number of genes likely contributing to class-specific differences.

Correlation between cancer and non-cancer: 0.946

Table 1: Descriptive statistics of dataset

Category	**Mean**	**Std**	**Min**	**Max**	**Median**
Overall	0	0.237	-1.302	5.515	-0.208
Cancer	-7.538×10^{-17}	0.217	-1.207	5.515	-0.246
Non-Cancer	9.422×10^{-18}	0.254	-1.302	5.486	-0.166

3.2 Histograms of statistics for all genes

The histograms of gene expression statistics reveal key patterns in the data. Most genes exhibit mean values clustered below zero, with long right tails indicating a subset of highly expressed genes in both tumor and non-tumor groups. The standard deviation peaks around 0.5-0.6, reflecting the expected variability in standardized data. Distributions of median and minimum values confirm the overall skewness, while maximum expression levels show pronounced right tails in both tissue types, highlighting potential biomarker candidates. These findings establish a basis for further analysis, particularly in identifying genes that best differentiate between cancerous and normal tissues.

3.3 Correlation between samples

To understand how similar or different the samples are in terms of their gene activity, we created heatmaps showing how strongly each sample correlates with the others.

The samples from healthy tissue show a lot of variety — many of them are not very similar to each other, as seen by the lighter colors on the heatmap. This means that gene expression in non-cancerous tissue can differ quite a bit from one person to another. These differences could be due to individual biological variation, lifestyle, or other non-cancer-related factors.

The cancer samples look more alike — they have stronger correlations with each other, which shows up as darker blocks on the heatmap. This suggests that once cancer develops, it tends to drive the cells into more similar patterns of gene activity. Even though tumors can be different in type or stage, they still seem to share common features that make their gene expression more consistent.

When we look at all the samples together, we can clearly see two groups forming — one for cancer and one for non-cancer. The samples within each group are more similar to each other, while cancer and non-cancer samples are quite different from one another. This clear separation suggests that cancer has a strong and noticeable effect on gene expression, making it possible to distinguish tumors from healthy tissue based on these patterns.

3.4 Most variable genes

To better understand which genes show the most differences across the samples, we plotted boxplots for the 10 most variable genes in the dataset. These are the genes whose expression levels change the most between different samples — potentially indicating important roles in distinguishing cancer from non-cancer tissue. The distribution of these genes is presented in Figure 3, which allows for a clear visual comparison of expression patterns across the two groups and highlights genes that may serve as potential biomarkers.

Each box represents the spread of expression values (after standardization) for one gene across all 102 samples:

Figure 1: Histograms of selected statistics

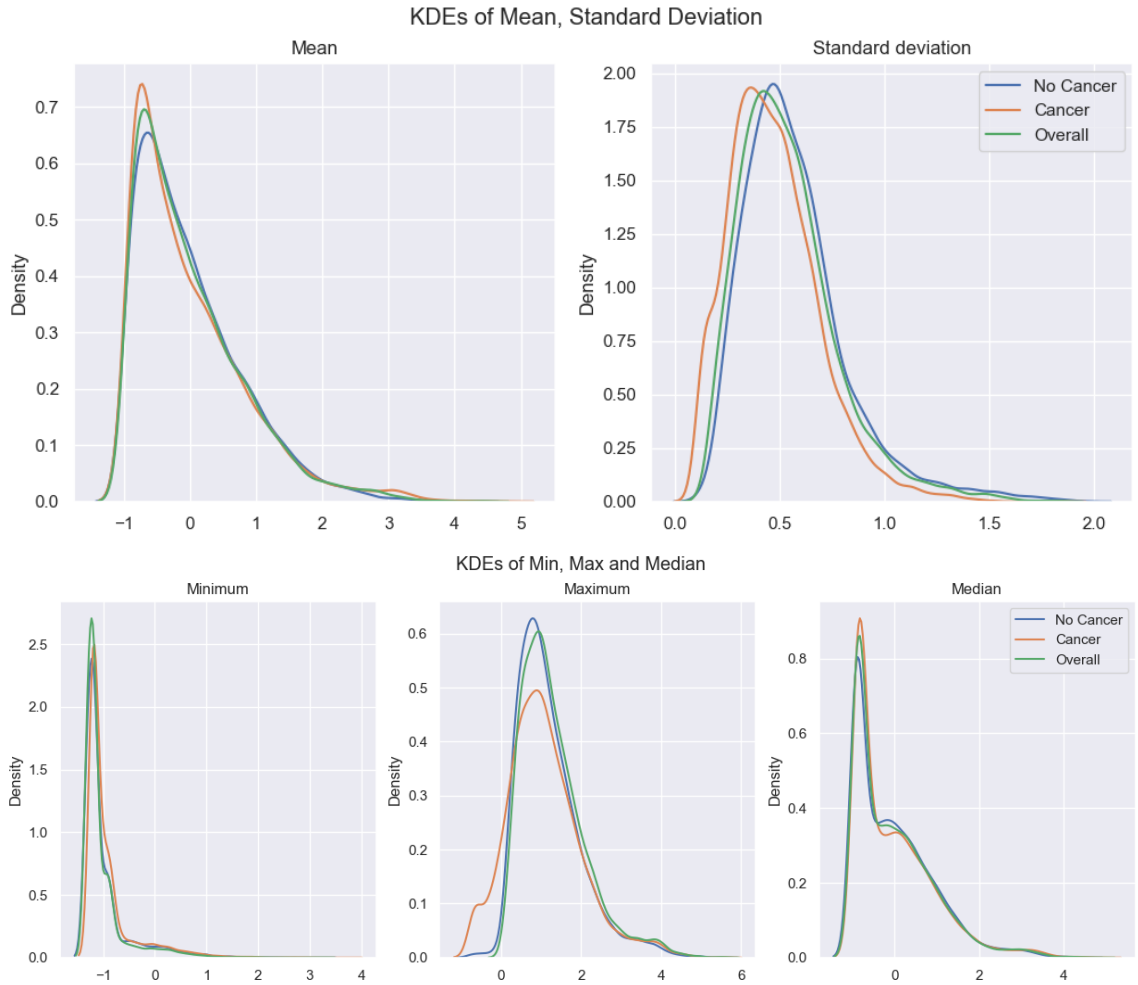
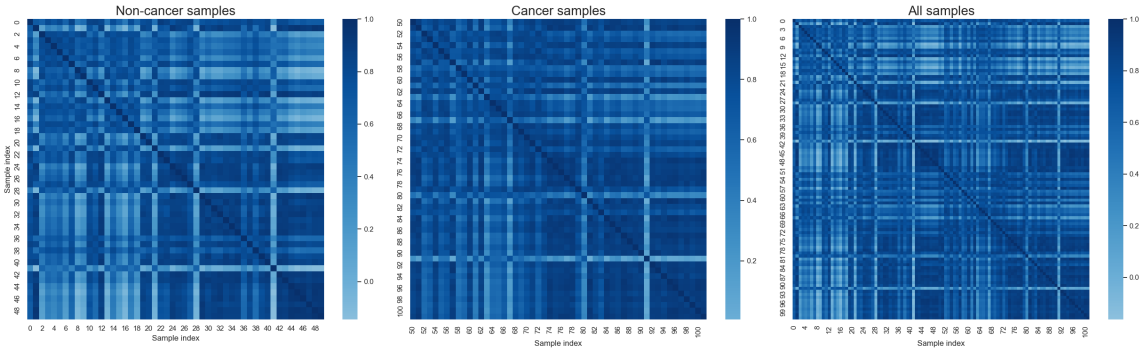


Figure 2: Correlation heatmaps



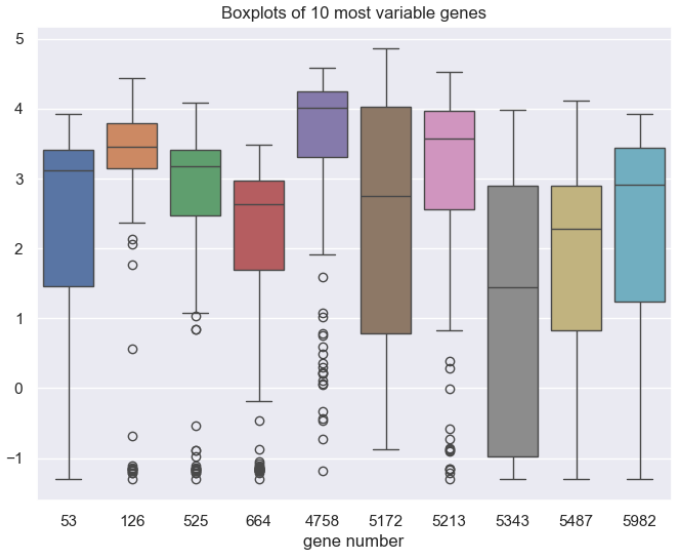
Some genes, like gene 4758 and gene 5172, show a wide range of expression values, meaning they vary a lot between samples. This high variability makes them potentially useful for classification tasks or further biological investigation.

Other genes, such as gene 126 and gene 5213, have tighter distributions with less variation, but still made it into the top 10 based on their overall variability.

Several genes (e.g. gene 5343 or gene 664) also show a noticeable number of outliers — individual samples with unusually high or low expression — which might reflect biological differences or technical variability.

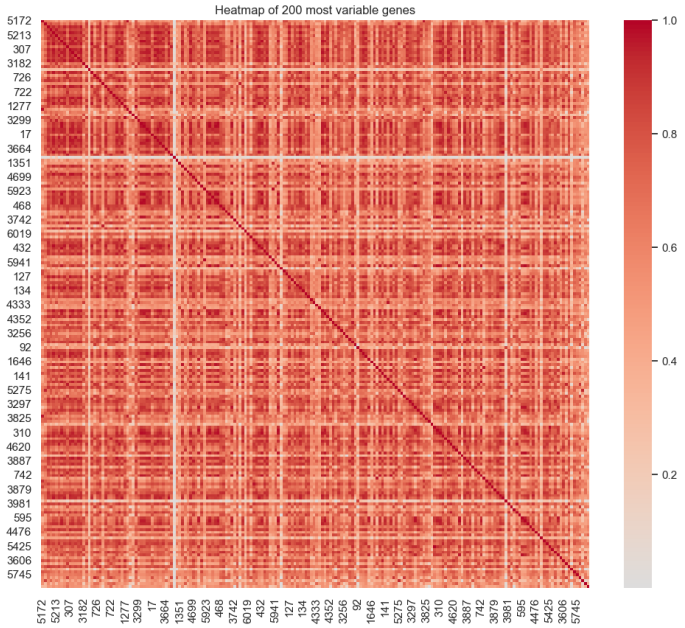
Overall, these highly variable genes are good candidates for identifying patterns or building predictive models, since they carry the most signal in terms of how expression differs between samples. The heatmap 4 shown above visualizes the correlation structure among the 200 most

Figure 3: Boxplots of top 10 most variable genes



variable genes across all prostate samples in the dataset. Each cell in the matrix represents the degree of correlation between a pair of genes, with darker red shades indicating stronger positive correlations, and lighter shades reflecting weaker associations. As expected, the diagonal is deeply red, corresponding to the perfect correlation of each gene with itself. However, what stands out are the numerous off-diagonal areas that also exhibit relatively high correlation values. This pattern suggests that certain genes tend to be co-expressed, possibly due to shared regulatory mechanisms or involvement in similar biological pathways. The grid-like structure observed in the heatmap implies the presence of gene clusters that behave similarly across samples, which could be particularly relevant in distinguishing tumor from non-tumor tissues or in identifying potential biomarkers for prostate cancer. This visualization provides an initial insight into the complex relationships between genes and serves as a foundation for further analysis, such as clustering or dimensionality reduction techniques.

Figure 4: Heatmap of top 200 most variable genes



3.5 Distribution of top variance features by cancer status

Figure 5 illustrates the distribution of the top-variance features in the dataset, split by cancer status (tumor vs. non-tumor samples). Each point on the plot represents a feature value for a specific sample, with red dots indicating cancerous tissue and blue dots representing non-cancerous tissue. The y-axis contains randomly jittered values purely for visualization purposes, allowing the points to be spread vertically and avoiding overlap.

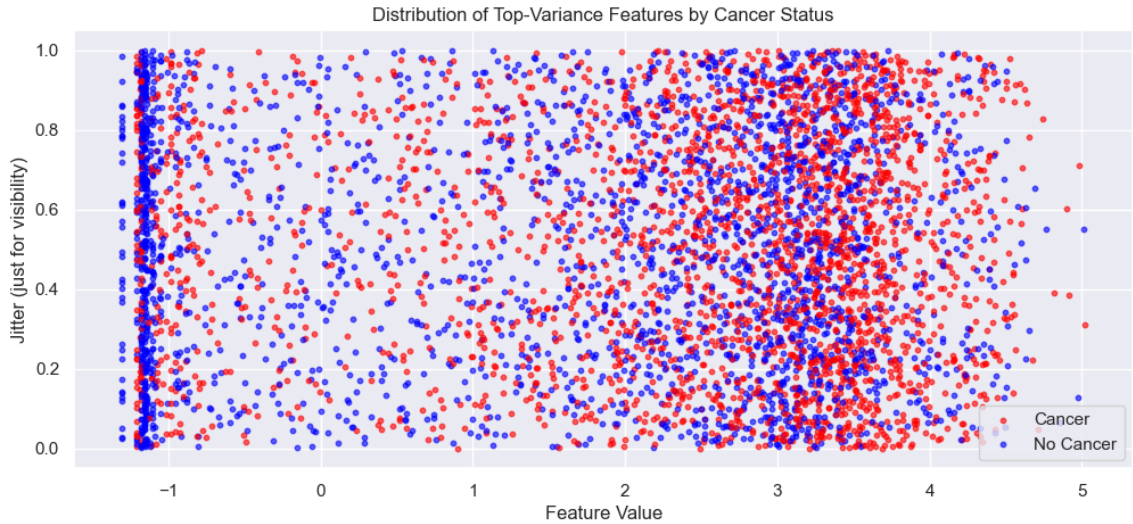
From the distribution, it is clear that both cancer and non-cancer samples span a similar range of feature values, particularly between 1.5 and 4.5 on the x-axis. However, the denser clustering of cancer samples in the higher range of values (around 3 and above) suggests that, for many of these high-variance genes, tumor tissues tend to exhibit elevated expression levels. Conversely, the large concentration of points near -1.5 to 0 may reflect standardized gene expression values close to or below the mean, with no strong separation between classes in that range.

Overall, while the separation between cancer and non-cancer samples is not starkly visible in individual features, the plot hints at underlying shifts in expression profiles that, when combined, could contribute to distinguishing between the two tissue types. This reinforces the need for multivariate approaches to effectively capture the complex patterns associated with cancer presence.

4 Feature Selection and classification

Due to the very high number of features in the microarray dataset — over 6,000 gene expression variables — this report avoids using pure wrapper methods for feature selection, as they are too computationally expensive. For example, selecting just 20 genes from 6,000 would involve evaluating

Figure 5: Distribution of top variance features by cancer status



more than 10^{58} combinations, which is clearly unfeasible. To deal with this, two hybrid methods are used that combine fast filtering with model-based refinement. The first uses a standard univariate test (ANOVA), and the second a non-parametric alternative (Wilcoxon rank-sum test). Both are followed by a sequential selection algorithm that adds genes one by one based on model performance. This approach helps find a small set of the most relevant genes in a practical and efficient way.

The implementation starts by splitting the data into training and test sets, keeping the class balance intact. Gene expression levels are grouped by cancer status, and the Wilcoxon test is run to compare them across the two groups. A Bonferroni correction is applied to reduce the risk of false positives, and only genes with corrected p-values below 0.05 are kept. These shortlisted genes are then passed to the Sequential Feature Selector, which picks the best combination of 20 genes using several classifiers — logistic regression, LDA, QDA, KNN, and SVM. Each model is evaluated using accuracy, F1 score, and ROC AUC, and the whole process is repeated for each metric and classifier. Results like confusion matrices and ROC curves are collected to allow detailed comparison.

4.1 Wilcoxon test

4.1.1 Feature selection

The table below presents the classification performance of five different machine learning models, evaluated after feature selection using the Wilcoxon rank-sum test and sequential feature selection.

The classification results indicate that Logistic Regression, Linear Discriminant Analysis (LDA), and Support Vector Machines (SVM) achieved the highest performance, with an accuracy of approximately 0.905, F1 score of 0.900, and ROC AUC up to 0.964. Notably, both Logistic Regression and LDA reached identical scores across all metrics, suggesting consistent model behavior and effective feature selection. In contrast, Quadratic Discriminant Analysis (QDA) showed reduced performance, particularly in accuracy (0.761) and F1 score (0.857), which may reflect its sensitivity to the distributional assumptions that are not well satisfied in the selected feature space.

Table 2: Descriptive statistics of dataset

	Accuracy	F1-score	ROC AUC
Log	0.905	0.900	0.891
LDA	0.905	0.900	0.945
QDA	0.857	0.706	0.682
KNN	0.857	0.900	0.891
SVM	0.905	0.900	0.891

K-Nearest Neighbors (KNN) performed comparably to the top models in accuracy and F1 score but achieved a lower ROC AUC (0.891), indicating less confidence in probability estimates despite correct classifications.

Overall, these results demonstrate that combining the Wilcoxon test with sequential feature selection yields a compact and discriminative gene subset that enables high classification performance across several standard machine learning models. Logistic Regression and LDA, in particular, appear to offer the most robust and generalizable results in this setting.

4.1.2 Classification

The following figures provide a graphical summary of the classification results, including ROC curves and confusion matrices, obtained after gene selection using the Wilcoxon test. These visualizations offer additional insight into the predictive performance and classification behavior of each model.

The classification results indicate that Logistic Regression, Linear Discriminant Analysis (LDA), and Support Vector Machines (SVM) achieved the highest performance, each with an accuracy of approximately 0.905, F1 score of 0.900, and ROC AUC reaching up to 0.964. The identical scores between Logistic Regression and LDA across all metrics underscore their consistent and reliable behavior, suggesting strong feature compatibility and model robustness. SVM matches this high level of performance, reinforcing its effectiveness on the dataset.

In contrast, Quadratic Discriminant Analysis (QDA) demonstrates reduced effectiveness, particularly in accuracy (0.761) and F1 score (0.857). These results likely reflect the model's sensitivity to violations in its underlying assumptions regarding the data distribution. While K-Nearest Neighbors (KNN) achieves similar accuracy and F1 score to the top performers, its lower ROC AUC of 0.891 indicates weaker probability estimates and less confidence in its predictions, despite correctly classifying most instances.

4.2 ANOVA

4.2.1 Feature selection

The table below presents the classification performance of five different machine learning models, evaluated after feature selection using the ANOVA and sequential feature selection.

The classification results demonstrate that Logistic Regression, LDA, and SVM achieved identical and consistently high scores across all three metrics, with accuracy and F1 score both at 0.905 and ROC AUC at 0.964. This consistency suggests robust model performance and a well-chosen subset of predictive genes. KNN matched their performance in accuracy and F1 score (0.905 and 0.900, respectively), but exhibited a slightly lower ROC AUC of 0.891, indicating reduced confidence in probability estimates. QDA performed notably worse, with an accuracy of 0.762, an F1

Figure 6: Confusion Matrices for Wilcoxon test

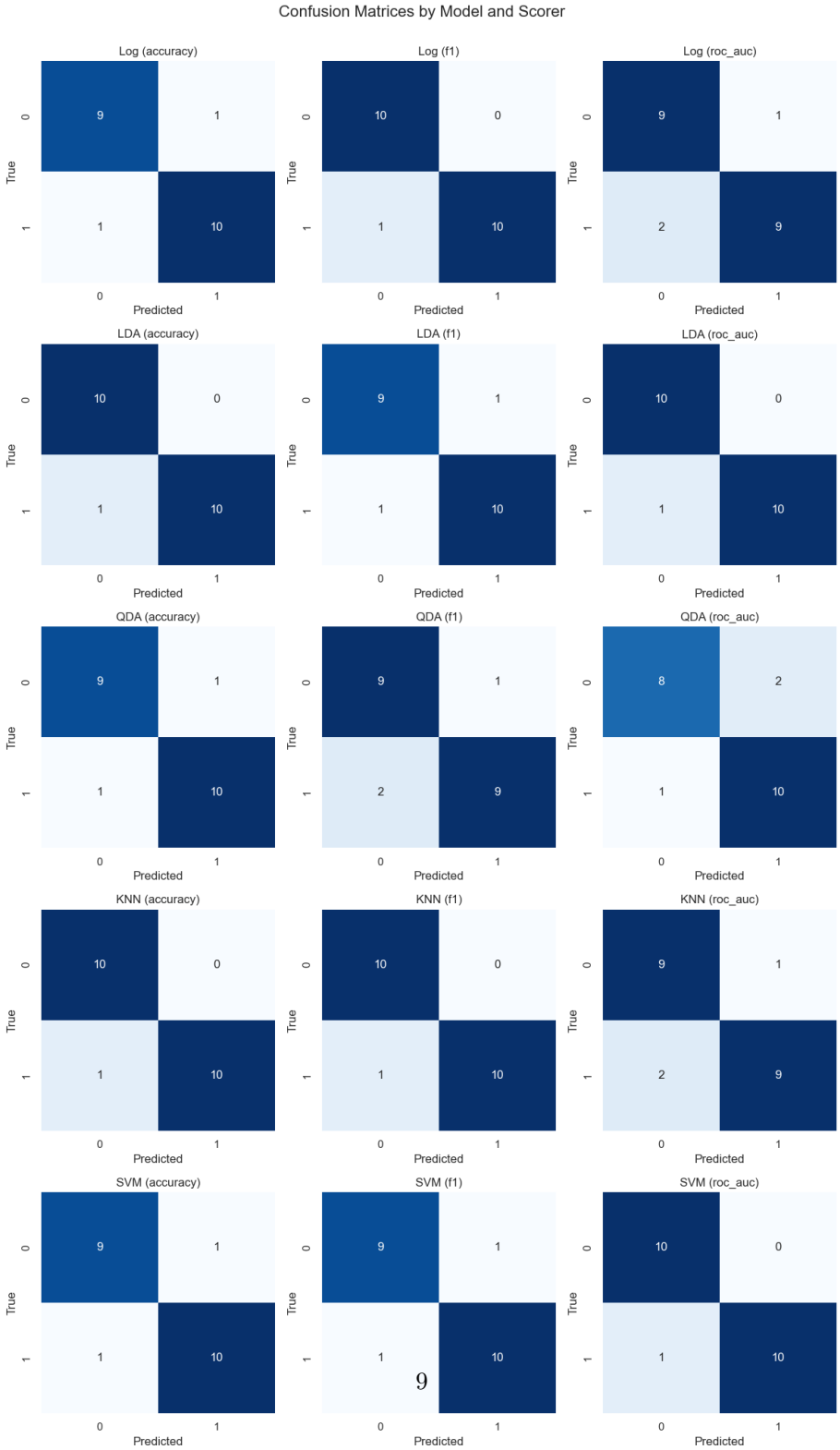


Figure 7: ROC Curves for Wilcoxon test

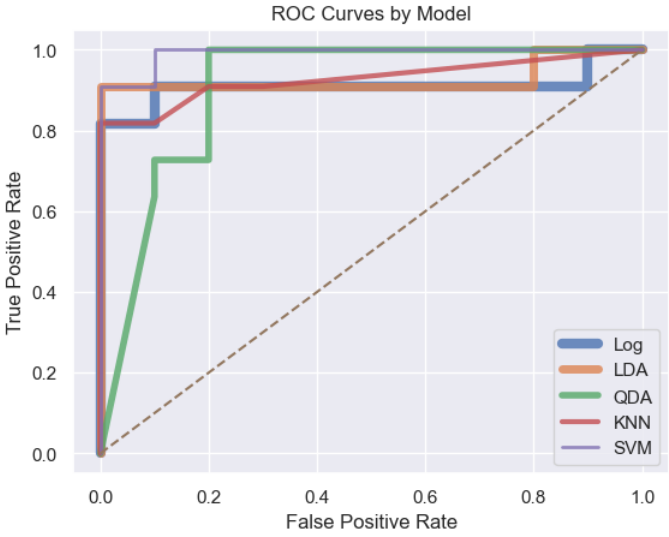


Table 3: Descriptive statistics of dataset

	Accuracy	F1-score	ROC AUC
Log	0.905	0.900	0.964
LDA	0.905	0.900	0.964
QDA	0.761	0.857	0.936
KNN	0.905	0.900	0.891
SVM	0.905	0.900	0.891

score of 0.857, and a ROC AUC of 0.936, suggesting that this model may be less suitable for the selected feature space, potentially due to its sensitivity to variance structure assumptions.

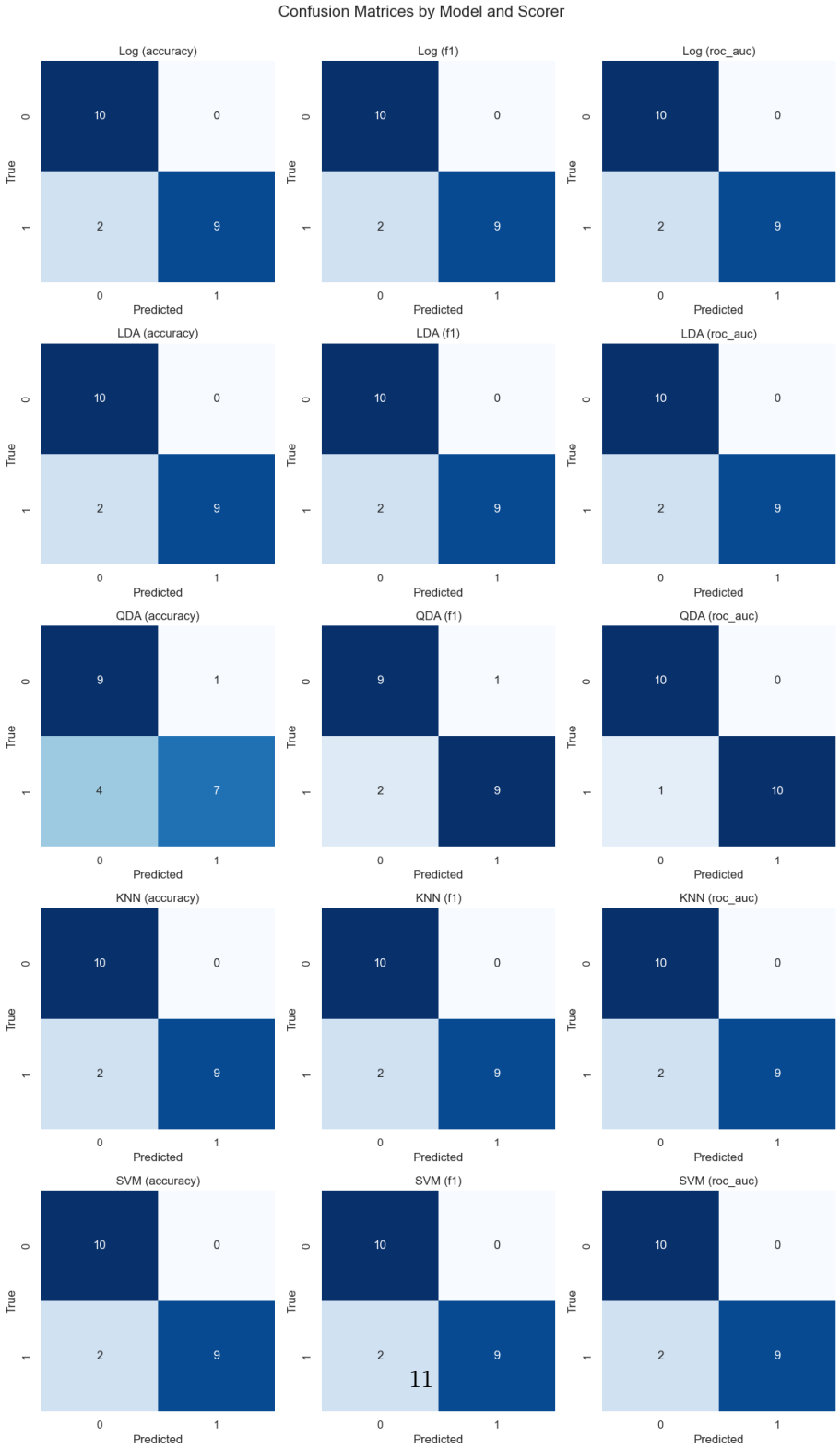
In summary, the ANOVA-based selection pipeline resulted in strong and stable classification performance for most models, particularly Logistic Regression, LDA, and SVM. These results confirm the effectiveness of combining univariate filtering with model-based refinement in high-dimensional settings.

4.2.2 Classification

The following figures provide a graphical summary of the classification results, including ROC curves and confusion matrices, obtained after gene selection using the ANOVA test. These visualizations offer additional insight into the predictive performance and classification behavior of each model.

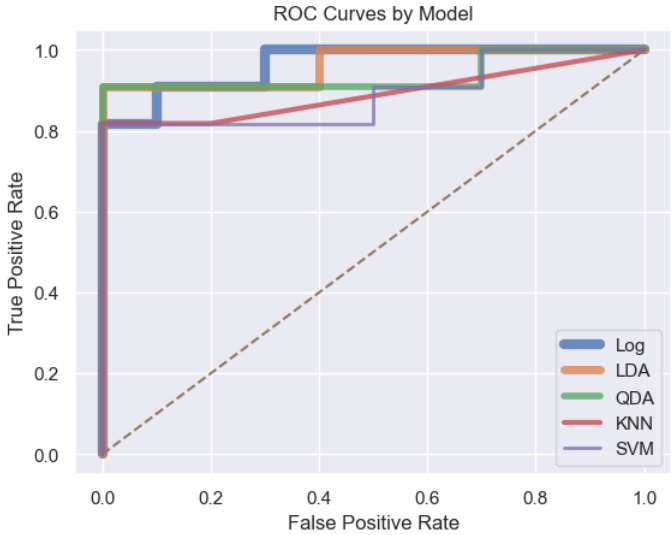
Confusion matrices reveal that Logistic Regression, LDA, SVM, and KNN all classified the data with high consistency, each making only two errors in identifying class 1. This suggests that these models handle the selected features well and provide stable predictions. In contrast, QDA shows a higher number of misclassifications, pointing to difficulties in adapting to the data's distribution. Overall, the confusion matrices highlight the strong and reliable performance of most models, with

Figure 8: Confusion Matrices for ANOVA



QDA standing out as less suitable.

Figure 9: ROC Curves for ANOVA



ROC curves further confirm the effectiveness of the top-performing models, with Logistic Regression, LDA, and SVM showing high separability between classes. Their ROC AUC scores are close to perfect, indicating strong confidence in their predictions. KNN performs slightly worse in this regard, despite similar accuracy, reflecting less reliable probability estimates. QDA, again, shows weaker performance, aligning with its results from the confusion matrices.

5 Conclusion

Although many individual genes in the prostate cancer dataset do not exhibit strong class-specific patterns on their own, when analyzed collectively, they reveal biologically meaningful structures essential for distinguishing between tumor and non-tumor samples. Features such as highly variable gene expressions and distinct correlation patterns across classes proved to be particularly informative. Despite the high dimensionality and subtle differences between the groups, descriptive analysis and visualizations uncovered consistent trends, clusters, and outliers that provided a strong rationale for classification efforts. The successful application of feature selection methods like the Wilcoxon test and ANOVA, followed by classification models, confirms that even in data with overall homogeneity, multivariate approaches can extract significant insights and yield robust predictive performance. This highlights the power of integrative data mining techniques in uncovering latent structure in complex biological datasets.